

ClientDeck: Product Specification & Development Blueprint

1. Executive Summary & Vision

ClientDeck is an open-source, self-hostable client portal designed to empower design agencies, software consultancies, and creative studios. Unlike generic project management tools, ClientDeck focuses exclusively on the high-stakes intersection of asset delivery and formal approval.

The vision is to provide a "Single Source of Truth" where agencies can present high-fidelity deliverables in a white-labeled environment. By centralizing the feedback loop and securing project sign-offs through an integrated audit trail, ClientDeck eliminates ambiguity and protects the agency's bottom line. As a self-hostable solution, it offers studios total data sovereignty and the ability to customize the portal to match their brand identity perfectly.

2. The Agency Problem

Modern creative workflows suffer from "Asset Sprawl." Valuable deliverables are scattered across fragmented channels:

- **Google Drive/Dropbox:** Final assets get buried in complex folder hierarchies.
- **Slack/Discord:** Feedback is lost in transient chat threads, leading to "I thought we agreed on this" disputes.
- **Email:** Versioning hell occurs as attachments fly back and forth with unclear naming conventions (e.g., `Final_v2_REVISED_real_final.png`).
- **Ambiguous Sign-offs:** Project milestones are often "approved" via a casual thumbs-up emoji, providing no legal or procedural weight when scope creep or payment disputes arise.

3. Core Feature Set

ClientDeck bridges the gap between creative work and business closure through a specialized feature set:

- **White-Label Dashboards:** Fully customizable client environments including custom logos, color palettes, and brand-specific subdomains.
- **Advanced File Versioning:** Upload multiple iterations of a single asset. The system automatically stacks versions, allowing clients to toggle between "Then" and "Now."
- **Canvas Annotation & Inline Commenting:** Visual assets (images, PDFs, and video) can be annotated directly. Clients can click any part of a design to leave specific, timestamped feedback.
- **Milestone Sign-off & Audit Trail:** Formalized "Approve" button for deliverables. Includes e-signature confirmation and a permanent log of who approved what and when.
- **Activity Feed:** A unified timeline showing all uploads, comments, and approvals for both the agency and the client.
- **Custom Domain Mapping:** Agencies can point `portal.agencyname.com` to their ClientDeck instance for a professional finish.

4. Developer Lead Responsibilities & Architecture

The Lead Developer  **Person** will oversee the technical implementation with a focus on scalability and ease of deployment.

Technical Stack

Layer	Technology
Frontend	React with Next.js (SSR for performance), Tailwind CSS for styling.
Backend	Node.js (TypeScript) with a PostgreSQL database.

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Storage	Object Storage integration (MinIO for self-hosted or AWS S3 for cloud).
Annotation	Custom HTML5 Canvas implementation or Fabric.js for vector overlays.
Infrastructure	Docker containerization for one-click deployment via Docker Compose.

Core Responsibilities

- **Architecture Design:** Implementing a multi-tenant-ready architecture within a single-tenant deployment model.
- **Permission System:** Building a robust Role-Based Access Control (RBAC) system to distinguish between Agency Admins, Project Managers, and Client Viewers.
- **API Security:** Ensuring all asset links are signed and time-limited to prevent unauthorized external access.

5. Business Lead Responsibilities

The Business Lead  **Person** will focus on the product-market fit and the open-source ecosystem.

- **Workflow Auditing:** Researching common agency friction points to refine the milestone approval UX.
- **Legal Template Integration:** Standardizing the language used in sign-off agreements to ensure they are professionally and legally robust.
- **Community Building:** Managing the GitHub repository, engaging with early adopters, and coordinating the launch on Product Hunt.
- **Demo Templates:** Creating "Project Starter Kits" for specific niches (e.g., a Branding Package template, a Web Dev Milestone template).

6. 4-Week MVP Roadmap

The goal for the MVP is a functional "Alpha" that can be used for a live project delivery by 📅 Date.

Week	Focus	Key Deliverables
Week 1	Foundation	User authentication, Database schema, Docker environment setup.
Week 2	Asset Management	File upload to S3/MinIO, Versioning logic, Basic dashboard UI.
Week 3	Feedback Loop	Canvas annotation tool, Comment threading, Activity notifications.
Week 4	Approval & Launch	Milestone sign-off module, E-signature logic, GitHub public release.

Distribution Model

ClientDeck will be distributed as an **Open-Source (AGPLv3)** project.

- **Community Edition:** Free to self-host with all core features.
- **Managed Hosting (Future):** A "Cloud" version for agencies who prefer not to manage their own servers, providing a recurring revenue stream for the project's sustainability.